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An Effective Hybrid Strategy: Multi-Fuzzy Genetic Tracking Controller for an Autonomous Delivery Van

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Abstract: The trend towards shorter supply chains and home delivery has rapidly increased delivery van traffic. Consequently, in the 20 years prior to 2018, delivery traffic has increased by 71%, while passenger vehicles have increased only by 13%. This drastic change in traffic patterns presented new challenges to decision makers and fortunately coincided with changes in the automotive industry, i.e., the advent of automation. However, the design of a controller is not straightforward due to the complex and nonlinear vehicle dynamics and the nonlinear relationship between the controller, tracking error and trajectory. This paper proposes a novel hybrid artificial-intelligence-based lateral control system for an autonomous delivery van to address these challenges to achieve the lowest value of tracking error. The strategy consists of multiple simultaneously operating fuzzy controllers. Their output signals are optimally weighted by a genetic algorithm to determine the proper allocation of control signals for calculating the final steering angle. Six different scenarios are implemented to evaluate the algorithm. A comparative analysis is then performed with two alternative state-of-the-art methods: (i) manually weighted and (ii) geometrically weighted controllers. During the tests, the vehicle's speed varied, and the roads considered ranged from simple roads to a series of curves. The results show that the proposed strategy leads to a reduction of up to 91.2% and 61.1% in tracking error compared to the manually and geometrically weighted alternatives, respectively.

Keywords: autonomous vehicle; tracking control; fuzzy logic; genetic; artificial intelligence; nature-inspired AI



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1. Introduction

In recent decades, intelligent transportation systems (ITSs) have attracted significant interest in improving road safety, as over 90% of all accidents are attributed to driver error [1]. Furthermore, traffic congestion caused by human and environmental protection are the other important motivating factors for the development of autonomous vehicles (AVs), as they have the potential to reduce fuel consumption (and thus, greenhouse gas emissions) by 45% [2]. Specifically, delivery traffic has increased by 71% in the 20 years prior to 2018, while car traffic has only increased by 13%, which in turn has led to a further increase in traffic [3]. In view of these considerations, the ultimate goal of the automotive industry is full vehicle autonomy [4]. However, ensuring safe operation is key to success

in promoting connected vehicles/AVs and is also a necessity for achieving the benefits of safety, mobility and sustainability [5].

1.1. Problem Statement

The three key enabling technologies for AVs are Sense, Plan and Act [6]. The third pillar, Act, deals with the tracking problem and is realised by control units to control either the vehicle's lateral or longitudinal motion or both simultaneously. Although there is an abundance of studies on conventional techniques [7–12], they reach their performance limits when the trajectory changes over time [13]. Consequently, research and development endeavours are increasing day by day, aiming at faster design, better performance and higher reliability. In this context, the central research question addressed in this study is as follows: How can advanced control methods be applied to optimise the tracking and safety of autonomous delivery vehicles under changing traffic scenarios?

1.2. State of the Art

Given the challenges posed by the dynamic and uncertain nature of real-world traffic scenarios, recent research has increasingly focused on advanced control strategies to enhance the safety and reliability of AVs. Furthermore, with increasing computational power, artificial intelligence algorithms have become a topic of interest in vehicle automation to improve performance [14–18]. One of these methods is fuzzy logic (FL) [19], which has been a prominent subject of interest in this field.

Since the 1980s, numerous fuzzy system applications have been reported in various scientific and engineering fields, particularly in automatic control, due to the fuzzy theory's suitability for handling nonlinear and complex systems, such as plant control [20–22]. One of the earliest applications of FL in AV control was demonstrated on a prototype in 1992 and successfully tested on a track [23]. Subsequent implementations on mobile robots, including a three-wheeled robot used in football tournaments and a four-wheeled robot tested for efficient FLC, have further validated the approach [24,25]. In [26], a robust fuzzy path tracker was designed and implemented using an experimental design technique based on the conicity stability criterion. The experimental results of the work demonstrated exemplary performance and robust behaviour in the presence of significant velocity variations. Similarly, a fuzzy-based longitudinal controller was developed in [27] and tested in the open racing car simulator environment. After automating the vehicle's shifting operations, an FL-based throttle/brake control system was developed, so that the race car was able to accelerate/decelerate in a realistic manner to drive at the desired speed. Unlike the studies using a single fuzzy architecture, an FLC consisting of multiple modules was also studied in [28] to control the steering of a two-axle vehicle towards its final destination while avoiding obstacles. A module to set the final alignment and a Bug steering module in case of failure were integrated with the control unit. Each effect on the final steering angle was analysed and adjusted according to the vehicle's environment [28]. The work was later extended via the integration of a fuzzy longitudinal controller, the simulation results of which revealed the successful operation of the vehicle in reaching the destination while bypassing all obstacles [29]. In [30], a model predictive observer was used together with an FLC. The observer was used to predict future states, while the fuzzy controller was used to control the vehicle. A comparative study was conducted using a PID and a traditional FLC. The predictive fuzzy controller performed better than its traditional fuzzy counterpart, which outperformed the PID controller. In [31], Yang and Zheng used an expert fuzzy lateral controller instead of a simple fuzzy control architecture. They designed an expert fuzzy lateral controller that can handle different driving speeds and curvatures for crossing and turning with a fixed look-ahead distance. Both the simulation and experimental results

showed the validity of the proposed approach. Multiple fuzzy systems have been proposed for complex AV operations, including parallel parking, T-S inference-based stability control and urban driving scenarios [32–35]. Wang et al. [36] further analysed the capabilities of multiple fuzzy systems, which are also considered within state-of-the-art controller in this paper. In particular, Wang et al. designed a new sensing system model and a descriptive model of the reference path used to estimate the lateral offset and heading angle error at single and multiple look-ahead distances. The proposed approach for lateral control was implemented on the IN2BOT autonomous vehicle and tested on two reference paths: a rounded rectangle and an S-shaped path. The results showed the effectiveness of the proposed control strategy under different driving conditions. Beyond architectural innovation, several studies have addressed FLC performance challenges, particularly in handling the nonlinearity of feedback tracking systems through alternative tuning strategies. These include rule generation via self-organising neural networks [37,38], reinforcement learning approaches, such as Q-learning [39], and population-based metaheuristic optimisation techniques, including genetic algorithms [40], particle swarm optimisation, ant colony optimisation and their hybrids [41–44]. These studies have shown varying degrees of success in improving inference accuracy and robustness. In a recent work, Liu et al. [45] used FL to adaptively tune a model predictive controller for AV path tracking. Liu and Shao [46] applied FL to enhance sliding mode control for yaw stability. These works show fuzzy control's evolving role in AVs.

1.3. Contribution

Even though most of the studies cited above dealt with nonlinear vehicle dynamics, they mostly overlooked the nonlinear relationship between the controller, the reference trajectory and the errors. However, there is a strong coupling between these constituents, necessitating their inclusion with optimum weighting and proper allocation to improve tracking performance in a broader range of operations without increasing the design effort, which is required by the rapid changes in the automotive industry. Therefore, this paper proposes a novel hybrid artificial-intelligence-based control system for an autonomous delivery van to address this challenge. In this regard, multi-fuzzy genetic control, fL_{GA} , is designed which consists of two fuzzy controllers considering the lateral and angular errors and reference input, whose outputs are then optimally weighted and allocated by a genetic algorithm (GA) to determine the final steering angle. The objective of the optimisation process is to obtain the lowest root mean square (RMS) value from the errors for the considered trajectory. With this hybrid approach, the design process becomes faster (with respect to the manually weighted method), more systematic and more robust due to the probabilistic transition rules that the GA inherently provides while being optimal [47]. The strategy is also evaluated against two alternative parameter tuning methods: (a) the manual method and (b) the geometric method [36] for comparative analysis.

The remaining sections are arranged as follows. Section 2.1 gives an overview of the vehicle model. Section 2.2 presents the proposed hybrid approach based on a multi-fuzzy genetic algorithm along with manually and geometrically weighted algorithms used for the comparative analysis. The performance indicators employed for evaluating all three strategies are presented in Section 2.3. In Section 3, the selected scenarios are presented, and simulation results are discussed in detail. Finally, the concluding remarks are given in Section 4.

2. System and Methodology

This section formulates the vehicle model and introduces the novel control strategy proposed in this study. Two rival control strategies are also defined for comparative

evaluation. Additionally, the performance metrics used for cross-evaluation of the control strategies are described.

2.1. Vehicle Model

This paper uses a nonlinear kinematic model to investigate controller performance for the proposed scenarios and speed range. First, a schematic representation of the bicycle model turning about its instantaneous centre of rotation (ICR) is illustrated in Figure 1, where L_f and L_r are the distances of the front and rear axles from the centre of gravity of the vehicle (CoG); δ is the steering angle of the front wheels; V_{CoG} is the velocity of the vehicle at the centre of gravity; and β_{CoG} and ψ are the vehicle sideslip angles and its heading angle with respect to the inertial coordinate X-axis.

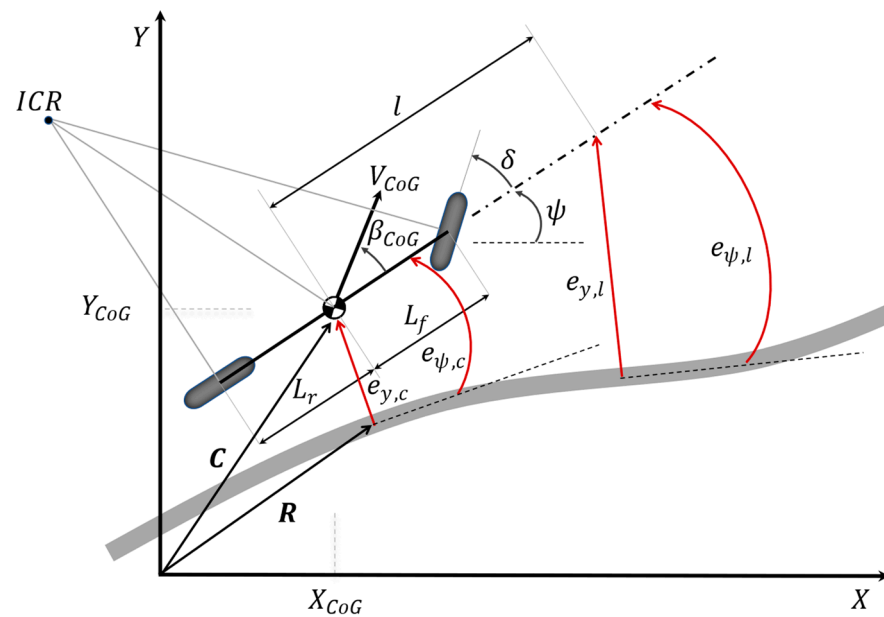


Figure 1. Nonlinear kinematic model of the Tofas-Fiat vehicle with a hypothetical reference path.

It is important to note that in the plant model, the sideslip is assumed to be zero. Therefore, the vehicle velocity at each tyre is aligned with the tyre direction. Equations (1)–(4) describe the governing equations for the CoG motion of the vehicle in the inertial frame ($X - Y$) [48], where X_{CoG} and Y_{CoG} are the vehicle CoG's coordinates. Throughout the analysis, CoG is assumed to overlap with the rear axle ($L_r = 0$) [49].

$$\dot{X}_{CoG} = V_{CoG} \cos(\beta_{CoG} + \psi) \quad (1)$$

$$\dot{Y}_{CoG} = V_{CoG} \sin(\beta_{CoG} + \psi) \quad (2)$$

$$\dot{\psi} = \frac{V_{CoG}}{L_r} \sin(\beta_{CoG}) \quad (3)$$

$$\beta_{CoG} = \tan^{-1} \left(\frac{L_r}{L_f + L_r} \tan \delta \right) \quad (4)$$

2.2. Artificial-Intelligence-Based Tracking Control Strategies

In this section, the theory underlying the developed lateral controller is presented. It also outlines state-of-the-art alternatives, which are used for comparison purposes.

2.2.1. Multi-Fuzzy Genetic Algorithm

This section presents a novel lateral tracking control paradigm for an autonomous delivery van. The strategy integrates a genetic algorithm with a fuzzy logic control system to improve the tracking performance of the reference trajectory. The hybrid strategy (fL_{GA}) therefore consists of dual FLCs that consider both the look-ahead lateral and angular errors: $e_{y,l}$ and $e_{\psi,l}$. The outputs of both FLC controllers are then optimally weighted by the GA algorithm considering the current lateral and angular errors $e_{y,c}$ and $e_{\psi,c}$ to determine the final steering angle.

The fuzzy rule base for each controller is constructed based on practical vehicle control principles. A total of 25 IF–THEN rules are used per controller. Triangular membership functions are assigned to all input and output variables, and the centroid method is used for defuzzification. The rules are designed to generate stronger corrective steering signals when tracking errors are large and the vehicle speed is low, allowing precise corrections under controllable conditions. Conversely, as the vehicle speed increases, the magnitude of the steering correction is intentionally reduced to prevent overshoot and oscillatory behaviour. Therefore, the maximum control signal is produced when the vehicle operates at a low speed with a high tracking error, while smaller corrections are issued at higher speeds, even in the presence of a significant error. Figure 2 illustrates the proposed strategy based on multiple fuzzy lateral controllers with GA-optimised weights, fL_{GA} . First, based on the vehicle model and the reference trajectory, the lateral look-ahead error, $e_{y,l}$, and the angular look-ahead error, $e_{\psi,l}$, are calculated. Each of these outputs is then fed into the position, and angular fuzzy controllers along with a secondary input which is derived from the velocity profile. Next, the fuzzification converts the crisp inputs into fuzzy variables $\hat{e}_{y,l}$ and $\hat{e}_{\psi,l}$. Accordingly, both controller outputs, $\hat{\delta}_y$ and $\hat{\delta}_\psi$, are processed by corresponding fuzzy inference systems. The fuzzified outputs are then subjected to a defuzzification process to obtain crisp values δ_y and δ_ψ .

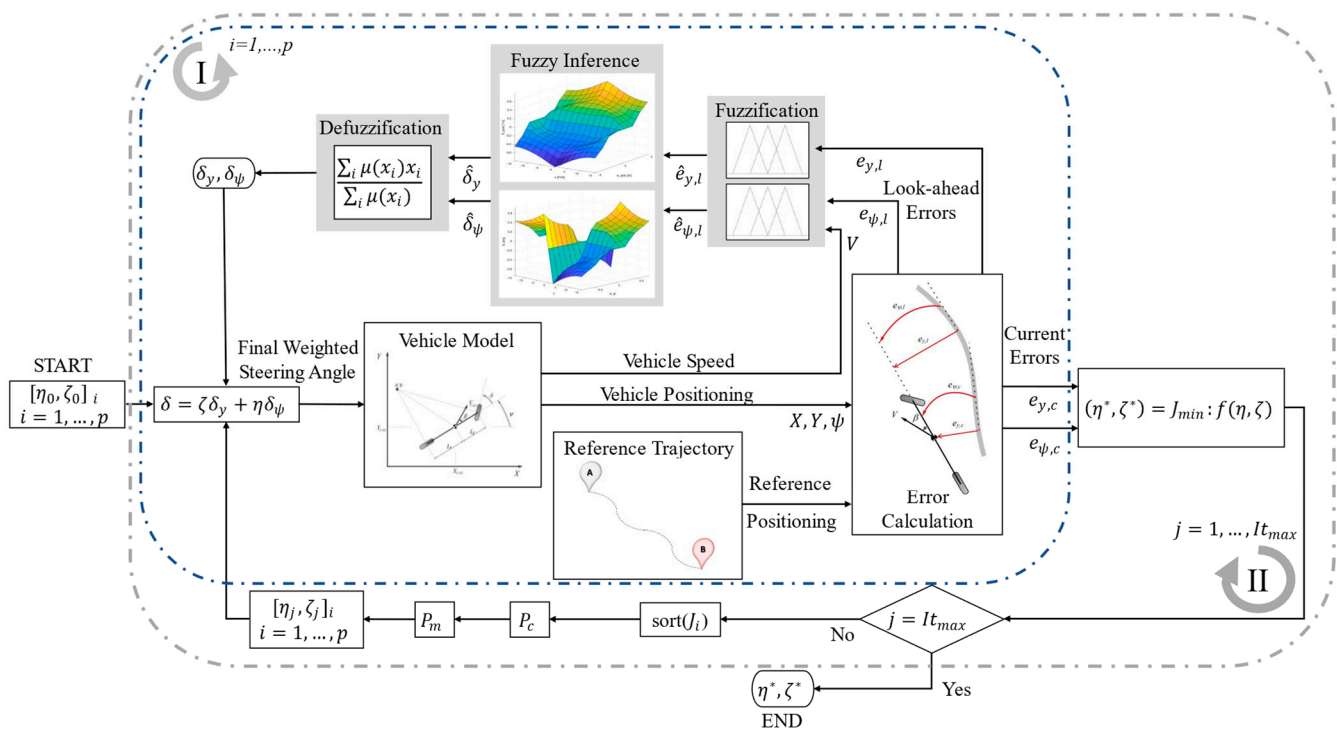


Figure 2. Schematic representation of the multi-fuzzy genetic controller (fL_{GA}) with the inner (I) and outer (II) optimisation layers.

To calculate the optimal steering input, δ^* , acting on the vehicle, the two output signals are multiplied by the optimal weights η and ζ , which are determined as a result of the genetic-algorithm-based optimisation process consisting of a main (II) and a subroutine (I). For the optimisation process, the cost function $f(\eta, \zeta)$ is chosen to minimise the RMS of the tracking error calculated based on the current lateral $e_{y,c}$ and heading angle error $e_{\psi,c}$ subject to the constraints given in Equation (5).

$$\begin{aligned} (\eta^*, \zeta^*) &= J_{min} : f(\eta, \zeta) \\ f(\eta, \zeta) &= \sqrt{\frac{1}{N} \sum_{j=1}^N (e_{y,c}(\eta, \zeta) + e_{\psi,c}(\eta, \zeta))^2} \\ &\text{subject to } 0 \leq \eta \leq 1 \quad 0 \leq \zeta \leq 1 \\ &\quad \eta + \zeta \leq 1 \end{aligned} \quad (5)$$

where N is the total number of samples; (η, ζ) is the lateral and heading steering angle weighting pairs; and (η^*, ζ^*) represents their corresponding optimum values.

The genetic algorithm used in this study is implemented using binary encoding with a resolution of 10 bits. The population size is set to 15, and the algorithm iterates over 50 generations. A roulette wheel selection method is employed to ensure diversity and fitness-based survival. The crossover operation is applied with a probability of 0.9 using one-point crossover, while mutation is performed by random bit-flipping with a mutation probability of 0.1 per bit. The best individual (elite solution) is preserved across generations to maintain optimisation progress. Each chromosome encodes a scalar weight that determines the influence of individual fuzzy controllers. These weights are used to combine the outputs of two fuzzy logic controllers: one responsible for lateral control and the other for angular control. Each fuzzy controller receives two inputs—the vehicle speed and the corresponding tracking error—and produces a local steering contribution. The final steering command is obtained via weighted combination of these two outputs, optimised via the genetic algorithm.

The optimisation process begins by selecting random initial populations, p , of η and ζ to obtain vectors η_i and ζ_i as lateral and angular weights, respectively. Next, the populations of η_i and ζ_i are combined with the initial lateral and angular steering angles to obtain the initial steering angle δ_i . The populations of η_i and ζ_i are then fed into the vehicle kinematic model to simulate the vehicle response and calculate the current lateral offset $e_{y,c}$ and the current heading angle error $e_{\psi,c}$. Then, the cost function is evaluated (Equation (5)). In this way, the algorithm sorts the population according to the corresponding tracking errors before the roulette wheel selection process starts, in which individuals are randomly selected. The selected individuals then undergo a crossover process in which they exchange characteristics. The mutation process then initiates abrupt changes in the population to avoid local optima by altering existing alleles through alternation. Both crossover and mutation are subject to probability satisfaction, P_c and P_m , respectively. Finally, the newly created individuals are returned to the vehicle model for evaluation, after which the process starts again until the maximum number of iterations is reached. The resulting outputs are the optimised parameters η^* and ζ^* , which are the inputs for the equation used to determine the optimised steering angle δ^* .

2.2.2. Alternative State-of-the-Art Tracking Control Strategies

For comparison purposes, two other state-of-the-art strategies are used: manually fL_M and geometrically weighted fL_{Geo} strategies.

fL_M : This baseline strategy is based on a single parameter, η , determined by the trial-and-error method with a constraint $\eta + \zeta = 1$. The process starts with the best estimate until the best possible performance is achieved. As expected, manual tuning of the weights is

tedious, time-consuming and subjective. Therefore, in the simulation studies, η is manually set to 0.85 after several iterations based on the expert's knowledge and perception. While this method provides a baseline for comparison, it is inherently subjective, time-consuming and not scalable across different scenarios or vehicles.

fL_{Geo} : Similar to the manual strategy fL_M , it is also based on an identical look-ahead error weight. However, the main difference of this comparative approach is that η ($= 1 - \zeta$) is dynamically determined by Equation (6) during trajectory tracking to reflect the importance of the trajectory through the road under consideration. This allows the weights to adapt to the curvature of the road and the anticipated trajectory changes, offering more flexibility and robustness than fL_M , especially under non-uniform driving conditions.

$$\zeta = 0.75 - 0.5E \quad (6)$$

where E is the evaluation function within the range of $[0, 1]$, as given in Equation (7)

$$E = \frac{1}{1 + \sum_{k=N_{now}}^{N_{far}} |x_k - (a_0 + a_1 y_k + a_2 y_k^2)|} \quad (7)$$

where N_{now} is the index of the point on the reference trajectory corresponding to the current location of the vehicle; N_{far} is the index of the first bending point on the local reference trajectory; (x_k, y_k) are the coordinates of the local trajectory; and $(a_0 + a_1 y_k + a_2 y_k^2)$ is the fitting curve of the local reference trajectory [36].

As shown in Table 1, a comparison is provided among the three control strategies evaluated in this study. This comparison framework highlights the added value of adaptivity of fL_{GA} . The table clarifies the superiority of each strategy and its limitations compared to the other two rivals.

Table 1. Overview of control strategy features for comparative analysis.

Controller	Weights Parameterisation Strategy	Superiority	Limitation	Objective Focus
fL_{GA}	GA optimisation	Optimally adaptable to any path	Pre-determination	Fixed and adaptable weights control performance
fL_{Geo}	Path-geometry-based	Relatively simple tuning	Not optimal	Varying weights baseline control
fL_M	Expert knowledge	Simple tuning	Not adaptable, poor performance	Fixed weights baseline control

2.3. Performance Indicators

Three key performance indicators are identified for evaluating the controller's performance in the selected scenarios:

(i) The root mean square (RMS) error between each current position and the corresponding reference point on the trajectory, given in Equation (8), is used to evaluate the overall accuracy of the controller's tracking performance. In other words, it measures the amount of deviation from the reference line points. The lower $e_{y,c,RMS}$ is, the better the tracking performance of the controller.

$$e_{y,c,RMS} = \sqrt{\frac{1}{N} \sum_{j=1}^N (e_{y,c})_j^2} \quad (8)$$

where the trajectory tracking error, $e_{y,c}$, is defined as

$$e_{y,c} = (-1)^d \|R(X_{CoG}(t), Y_{CoG}(t), \psi_{CoG}(t)) - C(X_{CoG}(t), Y_{CoG}(t), \psi_{CoG}(t))\|_2 \quad (9)$$

where C is the position vector of the CoG of the vehicle; R is the position vector of the point on the reference trajectory closest to the end point of C (corresponding to N_{now}); $x(t)$ is the horizontal coordinate; $y(t)$ is the vertical coordinate; $\psi(t)$ is the angular coordinate of the vehicle; and d is either 1 or 2 if the vehicle is on the left- or right-hand side of the reference path, respectively.

(ii) Maximum lateral error: The maximum lateral deviation from the path is also used to evaluate the controller's performance, as given in Equation (10).

$$e_{y,c,max} = \max(|(e_{y,c})_j|) \quad (10)$$

where $e_{y,c,max}$ is the maximum value of the CoG absolute lateral error along the trajectory.

(iii) Maximum heading angle error: The maximum angular deviation from the path is also used to evaluate the controller's performance, as given in Equation (11).

$$e_{\psi,c,max} = \max(|(e_{\psi,c})_j|) \quad (11)$$

where $e_{\psi,c,max}$ is the maximum absolute value of the vehicle heading angle error along the trajectory.

3. Results and Discussion

In this section, the performance of the proposed hybrid strategy, fL_{GA} , is evaluated against the other two methods: fL_M and fL_{Geo} . The results are compared using six scenarios simulated with the model of an autonomous delivery van. Figure 3 shows a simplified diagram of the overall simulation framework for the hypothetical Tofas-Fiat autonomous electric delivery van in MATLAB/Simulink (2022b version). The simulation framework consists of the following:

- i. The trajectory generation module defines the pose of the van ($x(t), y(t), \psi(t)$) considering the vehicle model to simulate six repetitive scenarios.
- ii. The look-ahead error blocks simultaneously compute the lateral offset and heading angle error with respect to the reference path, where the look-ahead distance is defined based on the fixed distance to mimic an expert driver who anticipates the distance he/she can travel in the city at low speed.
- iii. The control layer consists of the FL-based position and angle controller. The inputs of the controller are the velocity and the look-ahead lateral error $e_{y,l}$ and the look-ahead angular error $e_{\psi,l}$ and three blocks for setting the weights of η and ζ to determine the resultant steering angle δ^* .
- iv. The performance evaluation module is used for online assessment of the vehicle tracking performance for the selected strategy using the metrics specified in Equations (8), (10) and (11).

Six scenarios (S1–S6) are designed to evaluate the performance of the controllers. Three of the scenarios (S4–S6) are based on the main scenarios (S1–S3) involving different speeds, curves and manoeuvres. The results of the comparative simulation study on the three control strategies are analysed using two approaches to provide maximum insight into the performance of the proposed technique. The first approach consists of an illustrative analysis of the three main scenarios, for the sake of brevity. The second approach, however, is a quantitative comparative analysis of the performance metrics defined in Section 2.3 for all six scenarios. During the simulation studies, optimal η^* and ζ^* values are found

for all scenarios to optimise the controller outputs. The optimal weights that minimise the tracking error variance while satisfying the boundary condition of Equation (5) are given in Table 2. Based on the results, the largest η is obtained for Scenario 4 (S4) with a value of 0.98, while the opposite holds for Scenario (S1) with a value of 0.443, which is due to the complexity level of the trajectory and manoeuvre.

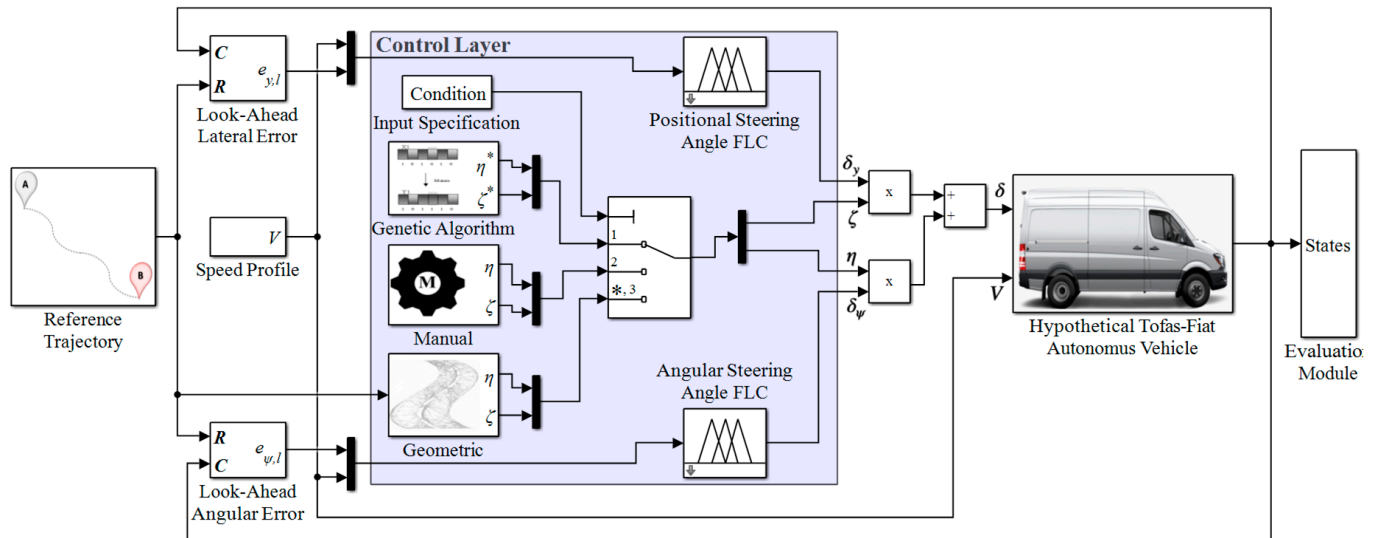


Figure 3. MATLAB/Simulink© comparative analysis framework.

Table 2. Optimal weights for the fL_{GA} strategy considered in different scenarios.

	S1	S2	S3	S4	S5	S6
η^*	0.443	0.449	0.881	0.980	0.456	0.449
ζ^*	0.557	0.551	0.119	0.020	0.544	0.551

3.1. Illustrative Analysis

Three scenarios are defined to perform the illustrative analysis of the comparative simulation study. Scenario 1 (S1) refers to a trajectory defined as a curved road with obstacles on which the vehicle must travel at a speed of 4 m/s (Figure 4a). The response of each strategy is analysed in terms of δ , $e_{y,c}$ and $e_{\psi,c}$ along the trajectory, as shown in Figure 4b–d.

Figure 4a shows the reference trajectory for S1, i.e., the centreline of the road, and the actual tracking performance of all three approaches, while Figure 4b–d show the vehicle steering angle, lateral offset error and vehicle heading error, respectively, at a speed of 4 m/s. Compared to the hybrid fL_{GA} and geometrically weighted fL_{Geo} strategies, the manually weighted fL_M applies maximum steering output during cornering ($t \approx 15$ s), peaking at -12.5° , indicating poor control ability (Figure 4b). Consistent with this, Figure 4c shows the lateral errors generated at the trajectory's first ($t \approx 4$ s) and second curvature ($t \approx 16$ s). As expected, the maximum error (oversteer) is generated by the manual-based algorithm, while the minimum error is generated by the genetic algorithm, proving that the proposed algorithm is able to significantly minimise the lateral errors at sharp curves compared to other tracking methods. Finally, in Figure 4d, fL_{GA} shows the best performance, while fL_{Geo} is the worst strategy in tracking the heading angle, with an error rate of over -6° at $t \approx 15$ s.

Similar to Figure 4a, Figure 5a illustrates the reference trajectory for Scenario 2 (S2), which is sharper and longer than S1 at a speed of 6 m/s, and the actual trajectory results of all three controllers. As a result, the actual trajectory of the vehicle with fL_{GA} is the

smoothest of the three, with a maximum steering angle of -14° at the sharpest point of the curve ($t \approx 11$ s), while the maximum steering angle of fL_M at the same curve is -16° (Figure 5b). Moreover, no noticeable oscillations occur until the end of the drive. Figure 5c shows the lateral errors corresponding to the three simulations. Accordingly, and consistent with Figure 5b, fL_{GA} shows the best performance with a maximum lateral error of -0.09 m, while fL_M shows the worst performance, and the lateral error after $t = 10$ s increases with time, reaching a maximum lateral error of 0.33 m at the end of the manoeuvre, while fL_{GA} settles to almost zero. For the heading angle errors, Figure 5d fL_{GA} yields -2.8° , while fL_{Geo} has the largest angular error peak of 4° at $t \approx 11$ s. As a result, fL_{GA} finds the optimal weight allocation among FLCs, providing the most uniform tracking performance among all other methods.

To further test the performance of the controller, simulation is performed with a more complex scenario, namely Scenario 3 (S3), which consists of a sinusoidal road, a U-curve and a double lane change followed by an overtaking manoeuvre (see Figure 6a).

As in the first two scenarios (S1 and S2), in S3, fL_{GA} exhibits the best tracking performance among the others (Figure 6a). Although the fL_{GA} and fL_M strategies have the same steering output, which is higher than that of fL_{Geo} , interestingly, fL_{Geo} shows an increasing time delay starting from $t \approx 3$ s, with few oscillations within the $t \approx 10 - 13$ s range (Figure 6b). Furthermore, fL_{Geo} has the highest lateral and heading angle errors, with more oscillations than the other two, while fL_{GA} has the lowest errors (Figure 6b,c).

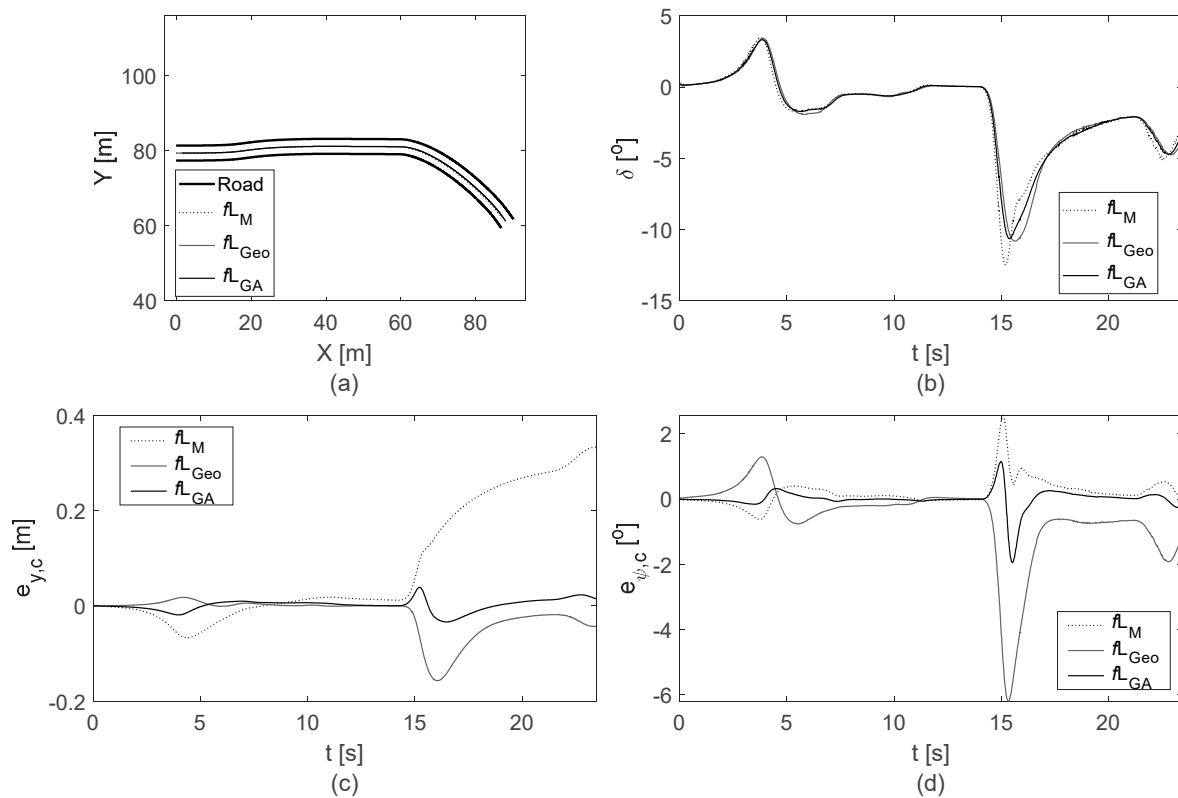


Figure 4. Vehicle (a) trajectory in the X-Y frame, (b) steering angle, δ , (c) lateral offset error $e_{y,c}$ and (d) heading angle error $e_{\psi,c}$ for S1 @ $V = 4$ m/s.

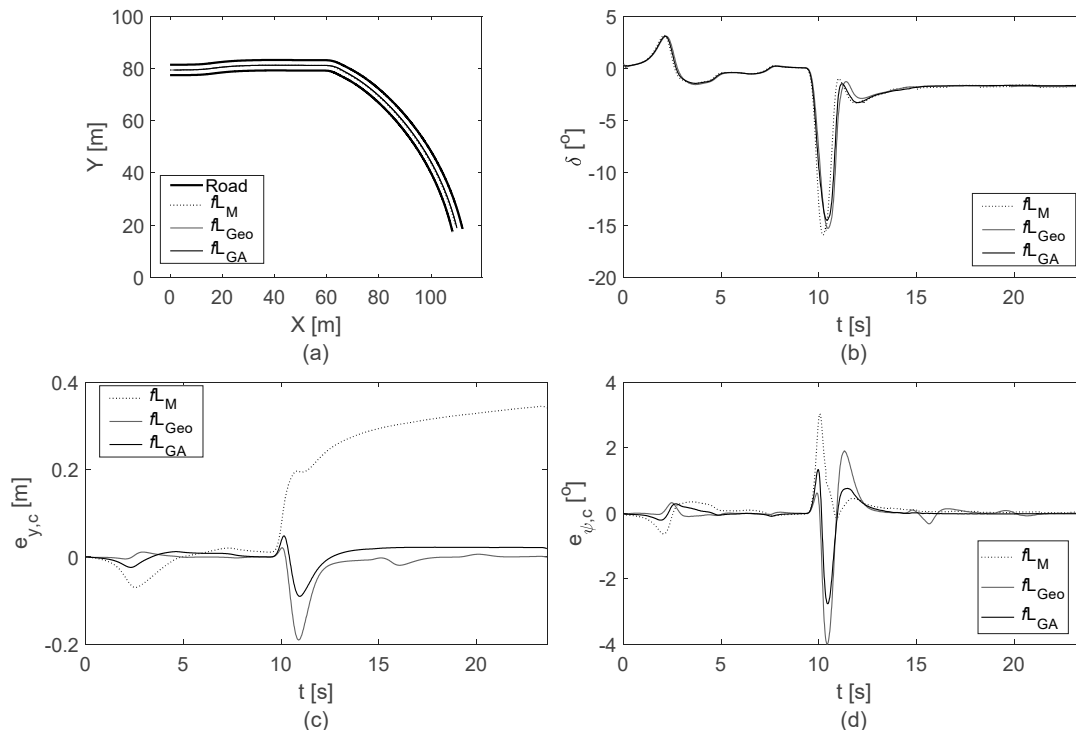


Figure 5. Vehicle (a) trajectory in the X-Y frame, (b) steering angle, δ , (c) lateral offset error $e_{y,c}$ and (d) heading angle error $e_{\psi,c}$ for S2 @ $V = 6$ m/s without obstacles.

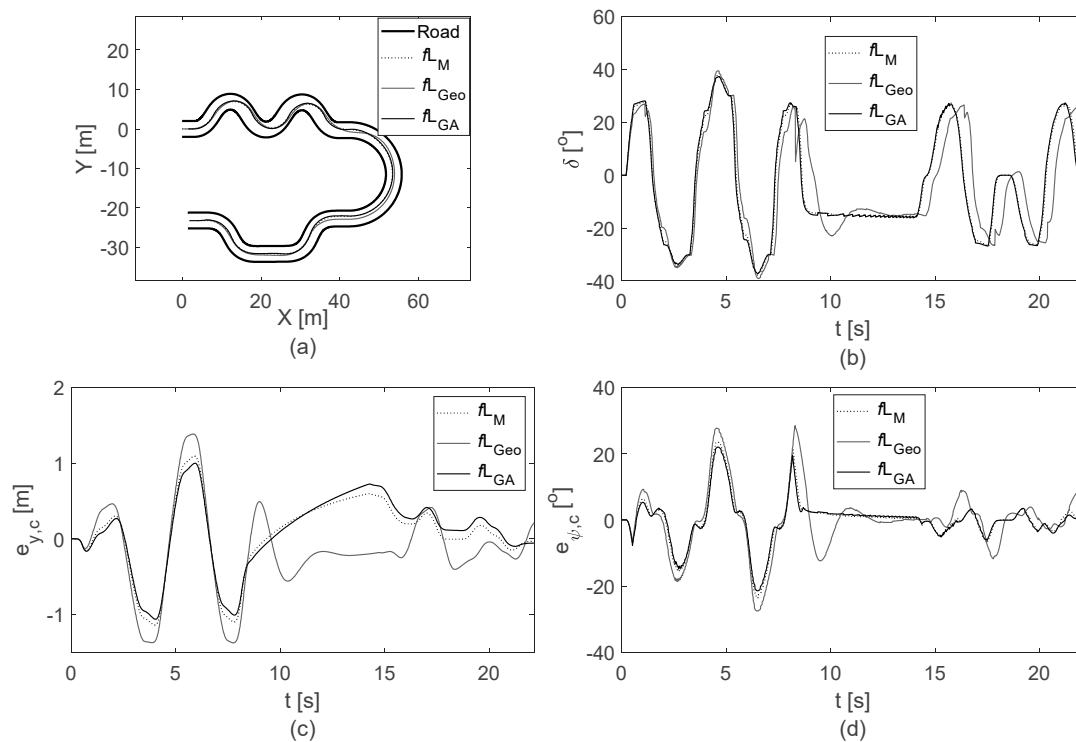


Figure 6. Vehicle (a) trajectory in the X-Y frame, (b) steering angle δ , (c) lateral offset error $e_{y,c}$ and (d) heading angle error $e_{\psi,c}$ for S3 @ $V = 6$ m/s.

3.2. Analysis Across Performance Metrics

Analysis through performance metrics is performed for the six scenarios, the first three of which are given in Section 3.1. The remaining scenarios are derivatives of the first three scenarios, with different manoeuvres and speeds.

Table 3 shows the root mean square tracking error based on Equation (8). The results show an average percentage improvement of 66.5% in lateral offset performance by fL_{GA} , with the slightest improvement at S3 being 1.2% and the largest at S5 being 91.2%. With fL_{Geo} , an average improvement of 44% is achieved, with a minimum of 17.1% for S3 and a maximum of 61.1% for S5. The results also show that fL_M performs the worst in terms of lateral error for all scenarios.

Table 3. Root mean square of the tracking error for each strategy considered in different scenarios.

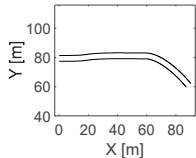
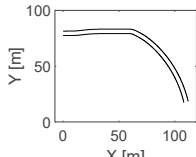
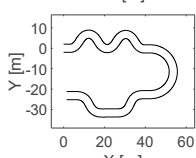
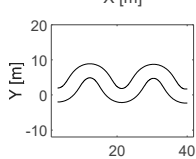
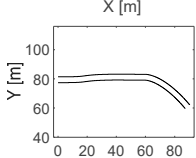
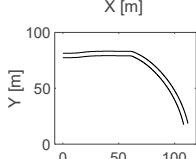
Scenarios		$e_{y,c,RMS} [m]$		
		fL_{GA}	fL_M	fL_{Geo}
S1		0.0138	0.154	0.0345
S2		0.0240	0.227	0.0398
S3		0.505	0.511	0.609
S4		0.501	0.793	0.989
S5		0.0132	0.150	0.0339
S6		0.0234	0.224	0.0369

Table 4 shows the maximum absolute lateral $e_{y,c,max}$ and angular $e_{\psi,c,max}$ errors determined using Equations (10) and (11) for all three control strategies and all six scenarios (S1–S6). The data in Table 4 show that fL_{GA} has the lowest lateral and angular errors, with average improvements of 56.1% and 18.2%, respectively, compared to fL_M and 45.9% and 44.4% compared to fL_{Geo} in all scenarios.

The proposed hybrid strategy leverages the strengths of a genetic algorithm to dynamically and optimally allocate weights between multiple fuzzy controllers. The objective function (defined in Section 2.2.1) is explicitly designed to minimise the RMS of the tracking error by using both current and look-ahead lateral and heading angle errors, subject to constraints defined in Equation (5). This enables the strategy to finely tune the control parameters for each scenario rather than relying on local or heuristic adjustments. Fur-

thermore, the superiority of the proposed controller stems from the inherent exploratory power of the GA, which effectively searches the solution space for optimal combinations of controller outputs. This is particularly advantageous when dealing with nonlinear, multi-objective or constrained optimisation problems, such as vehicle trajectory tracking, where analytical solutions are infeasible.

Table 4. Maximum lateral and angular tracking errors for each strategy considered in different scenarios.

Scenarios	fL_{GA}		fL_M		fL_{Geo}	
	$e_{y,c,max}$ [m]	$e_{\psi,c,max}$ [°]	$e_{y,c,max}$ [m]	$e_{\psi,c,max}$ [°]	$e_{y,c,max}$ [m]	$e_{\psi,c,max}$ [°]
S1	0.039	1.95	0.33	2.56	0.16	6.20
S2	0.090	2.76	0.35	3.04	0.19	4.00
S3	1.06	22.0	1.14	23.5	1.38	28.5
S4	0.76	15.9	1.15	23.9	1.41	28.0
S5	0.12	1.86	0.29	2.56	0.16	6.20
S6	0.090	2.76	0.35	3.04	0.19	4.00

4. Conclusions

The age of autonomous vehicles is approaching faster than we think as technology continues to evolve. The aim is to ensure road safety, reduce fuel consumption and mitigate the impact of greenhouse gas emissions with a holistic approach. One of the main pillars of vehicle automation rests on the tracking controller. This study addressed the following research question: How can advanced control methods be applied to optimise the tracking and safety of autonomous delivery vehicles under changing traffic scenarios? Although multiple controller-based strategies are able to take on different errors, as the complexity of the reference trajectory increases, the nonlinear relationship between the controller, the reference input and the errors also increases. Therefore, the weighted distribution and proper allocation of controller outputs become increasingly essential to achieve better performance. So far, the solutions offered by researchers have shown little demonstrable impact in addressing this problem. The idea is to define a novel controller based on a systematic approach that provides more robust and faster optimisation of constrained weights than two fine-tuning state-of-the-art alternatives to perform well even in challenging situations (see Table 4). In this context, this paper presents a novel paradigm for trajectory tracking control based on a hybrid approach that integrates fuzzy and genetic techniques, namely multi-fuzzy genetics fL_{GA} . The strategy uses current and look-ahead lateral offset and heading angle errors to compute the control inputs. At the same time, the genetic algorithm performs the optimal allocation of outputs of multiple fuzzy controllers to achieve the most uniform tracking performance, even for challenging trajectories. To evaluate the performance of the hybrid strategy fL_{GA} , both illustrative and quantitative analyses are performed using performance metrics. The significant improvement achieved by the controller compared to strategies where weights are adjusted manually fL_M and geometrically fL_{Geo} is shown via six scenarios. On average, of all scenarios, the RMS of the tracking error of fL_{GA} is 66.5% and 44% lower than that of fL_M and fL_{Geo} , respectively, with the most considerable corresponding differences being 91.2% and 61.1%. The importance of the proposed controller is also evident in the curves (S3), which resemble urban driving. Hence, the method is suitable for predefined routes, e.g., delivery vans, and it can be transferred easily to long-haul trucks. Moreover, using data maps can give the approach a broader application due to the improved real-time capabilities. Routes repeatedly travelled by autonomous delivery vans can benefit from this approach through the use of pre-optimised controller weights η and ζ . To facilitate this, the expected delivery routes can be discretised into representative driving scenarios. For each scenario, the methodology proposed in this study can be applied to derive optimal control

weights. These weights can then be stored in a database and retrieved in real time as soon as the vehicle's GPS confirms entry into a corresponding scenario. This strategy enables fleet operators to ensure consistent and seamless performance across diverse delivery routes.

While the proposed optimisation framework demonstrates significant improvements in terms of controller performance, it involves a computationally intensive tuning process due to the simultaneous optimisation of multiple parameters. The gain scheduling approach employed provides a flexible and effective way to manage controller outputs, though it can introduce additional layers of complexity that require careful handling. Future research could focus on streamlining the optimisation procedure to enhance computational efficiency to further improve real-time applicability.

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